

Team10

HelioBox







Have you ever seen a crack or pothole while driving?

Did you report it to the city immediately?

If you see one tomorrow... Will you report it?

Poor road conditions cost U.S. drivers billions annually

AAA: Potholes Pack a Punch as Drivers Pay \$26.5 Billion in Related Vehicle Repairs

New data shows 2021 was an expensive year for drivers who sustained vehicle damage from potholes



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ORLANDO, Fla. (Mar. 1, 2022)—Americans battling the polar vortex will soon face a greater enemy—pothole season. A new survey from AAA found that last year 1 in 10 drivers sustained vehicle damage significant enough to warrant a repair after hitting a pothole. With an average price tag of almost \$600 per repair, damage caused by potholes cost drivers a staggering \$26.5 billion in 2021 alone.

\$26.5B

Annual Cost

U.S. drivers spent \$26.5 billion on pothole damage in 2021.

1 in 10

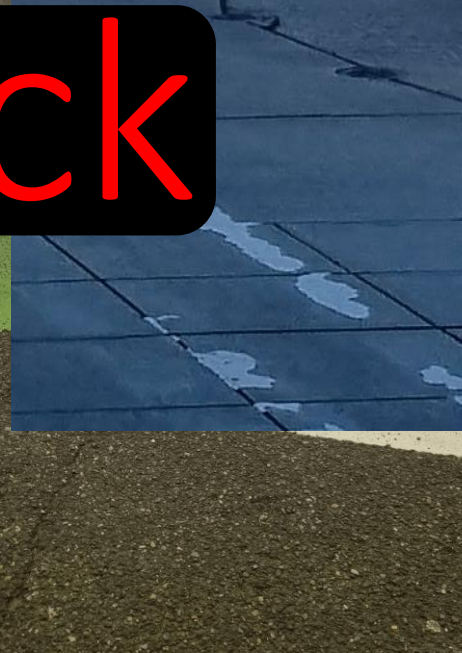
Drivers Affected

Drivers sustain vehicle damage annually due to poor road conditions.

So we search



A lot of crack



So the driver

Our Persona

The Interview of the Driver

"The city's response is always too late"

Insights from a Seattle driver struggling with frequent potholes

Pain Point

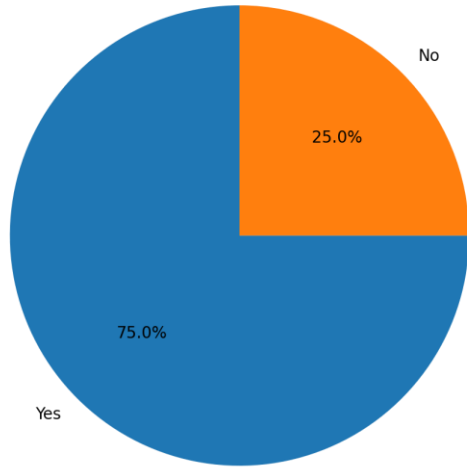
Personal frustration and mounting vehicle repair costs from repeated pothole damage

Reality

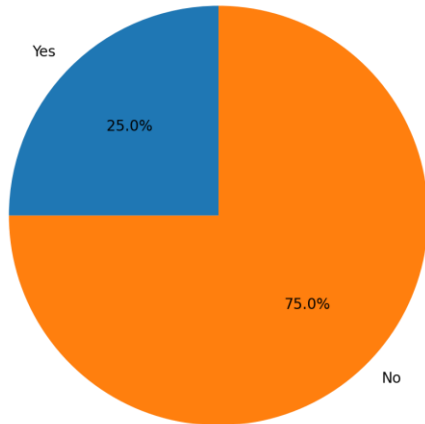
Current reporting systems are reactive and inefficient. By the time the city responds, multiple drivers have already suffered damage to their vehicles.



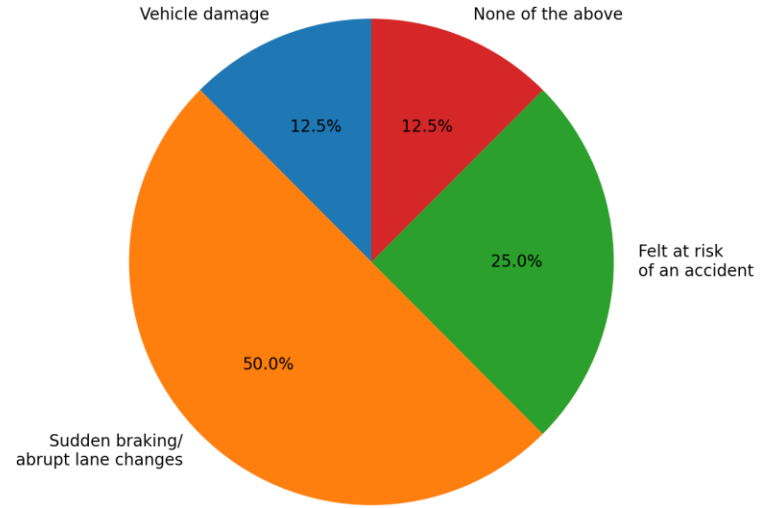
Q1 Experienced Inconvenience or Danger



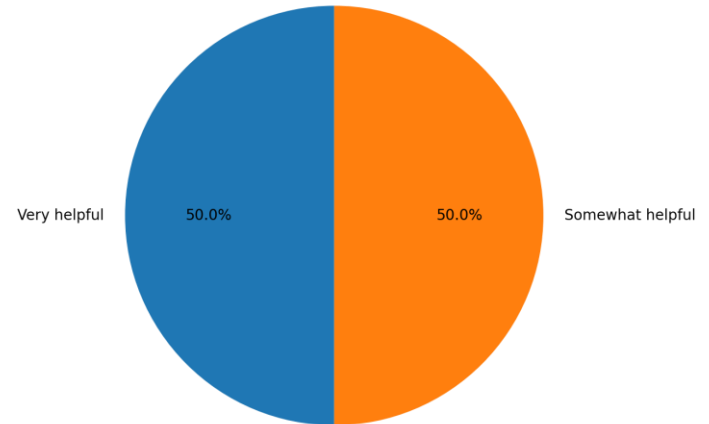
Q3 Reported to the City



Q2 Type of Inconvenience or Danger



Q4 Auto Detection System Helpfulness



The Rule of 1:10 – Timing is Money

\$1

Preservation

Early detection allows for cheap, preventive maintenance.

\$10

Rehabilitation

Reactive maintenance after a pothole forms is ten times more expensive.



Current manual inspections miss the critical \$1 opportunity.

So our Approach

We need specialized tools to monitor hazardous cracks continuously

1 Move from Reactive to Proactive

To prevent accidents, we must shift from responding to damage after it occurs to identifying risks before they escalate.

2 Continuous Data Collection

Real-time monitoring is the key to identifying structural weaknesses early, when they are still inexpensive to repair.

3 Make the Invisible Visible

Our goal is to transform hidden infrastructure weaknesses into actionable intelligence for city planners and maintenance teams.

Vehicle-Level Data Collection

RGB Camera System

Front windshield-mounted high-resolution camera capturing continuous video stream of road surface at 30 fps

GNSS/GPS Receiver

Precise location tracking with sub-meter accuracy and synchronized timestamps for each frame

Local Storage

SD/SSD card for buffering raw video and metadata during periods of limited connectivity

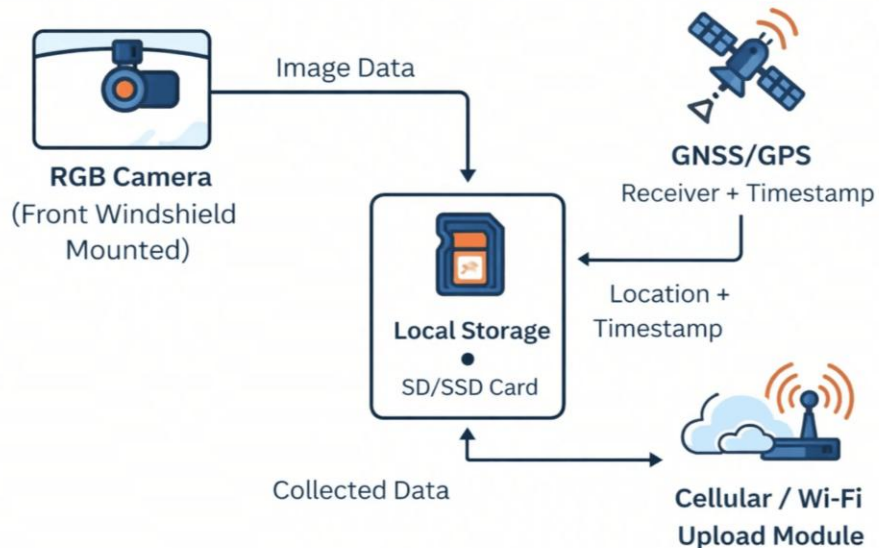
Connectivity Module

Cellular (4G/5G) or Wi-Fi for real-time data transmission to cloud infrastructure

Key Reference

Rocha et al. (2023). A Modular In-Vehicle C-ITS Architecture for Sensor Data Collection. IEEE Transactions on Intelligent Transportation Systems.

Vehicle-Level Data Collection for Road Crack Detection



Server-Level Processing

Stage 1

Frame Alignment & Preprocessing

Stabilize video frames and enhance image quality for analysis

Stage 2

Road Segment Division

Partition road into location-based segments using GPS coordinates

Stage 3

Crack Extraction (CV/ML)

Apply CNN models to identify crack candidate regions with high precision

Stage 4

Data Accumulation

Aggregate and compare multiple detections from same road segment

Stage 5

Confidence Scoring

Generate reliability scores for each detection result

Server-Level Data Processing in Road Crack Detection System



Key References

Zhang et al. (2022). Pavement Distress Detection using CNN. International Journal of Pavement Research. | Hu et al. (2025). Lightweight Deep Learning for Real-time Road Distress Detection. Nature Communications.

Fleet-Level Integration

Data Aggregation

Collect repeated route data from multiple vehicles across the fleet

Data Fusion

Integrate multi-vehicle observations for the same road segments

False Positive Removal

Apply statistical filtering to eliminate spurious detections

Confidence Scoring

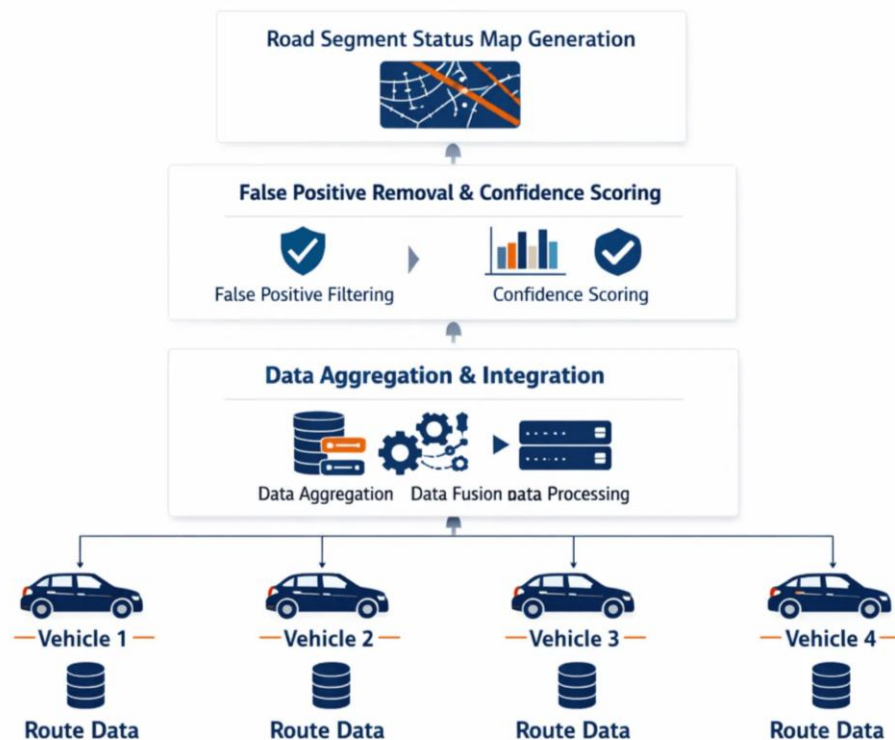
Assign reliability scores based on detection consensus across vehicles

Output: Road Status Map

Generate comprehensive city-wide infrastructure condition maps with high confidence scores

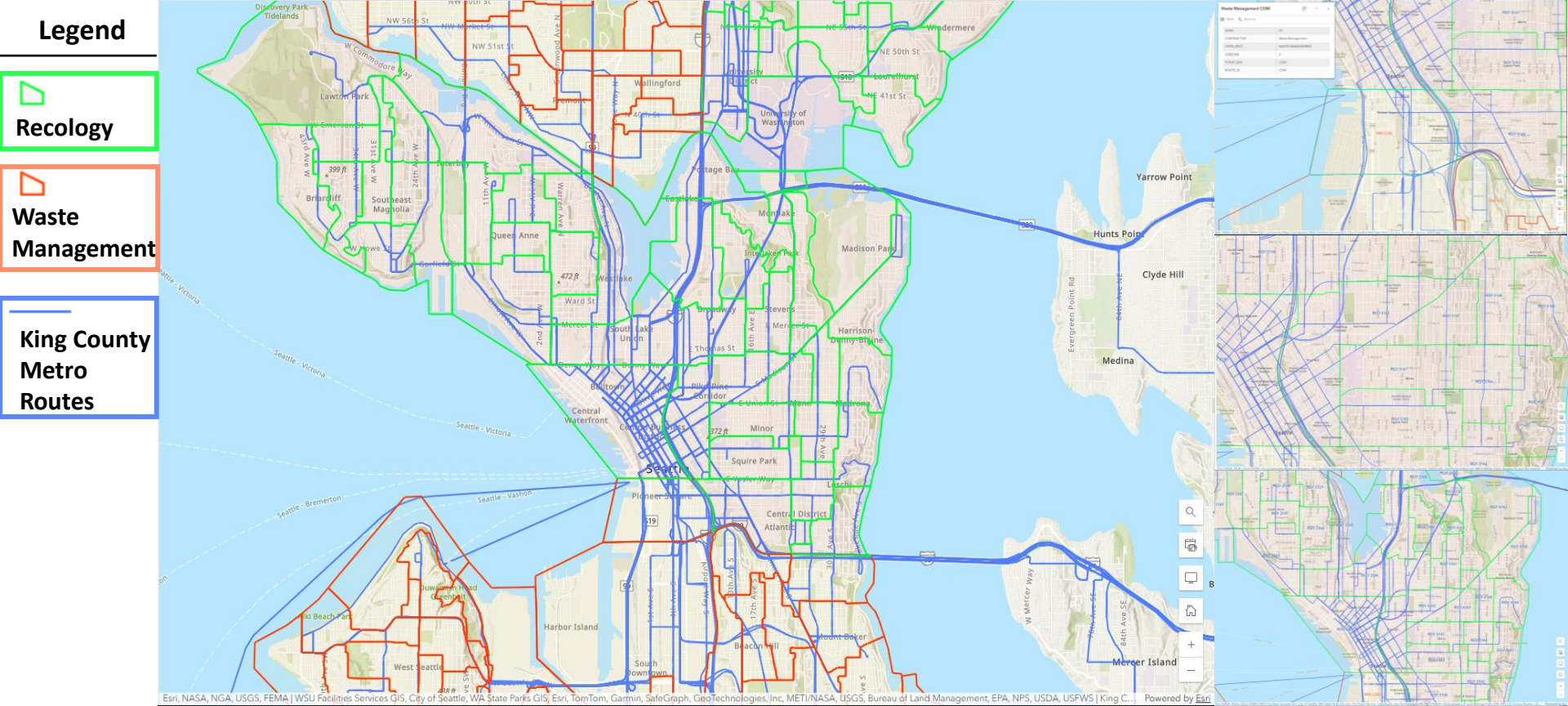
Key References

Rodrigues et al. (2011). A Mobile Sensing Architecture for Massive Urban Scanning. IEEE Conference on Intelligent Transportation Systems. | Chen et al. (2023). Computer Vision and Machine Learning for Road Surface Defect Detection. Journal of Smart Cities.



Conventional vs. Proposed Approach

Category	Conventional Approach	Proposed Approach
Target	Already-formed potholes	Road cracks at pre-pothole stage
Sensors	High-cost LiDAR, IMU, accelerometers	Commodity camera + GNSS location data
Processing	On-vehicle real-time processing	Server-based post-processing
Scalability	Per-vehicle standalone accuracy	Fleet-based cumulative accuracy improvement
Coverage	Limited coverage	Coverage scales with number of vehicles
Maintenance Cost	High	Relatively low



This image shows the waste management routes in the Bellevue and Seattle areas, visualizing designated zones for recycling and waste collection.



Leveraging city fleets for comprehensive road scanning

Layer 1

Speed Coverage

City buses scan main roads frequently for rapid updates.



Layer 2

Detail Coverage

Garbage trucks cover every single alley and side street.

Result

This approach allows us to cover most of the city without requiring additional personnel.

DEMO

D00

Detects vertical cracks running parallel to the road direction

D01

Identifies vertical joints or cracks along the road.

D20

Detects web-patterned surface damage, indicating structural fatigue in the pavement.

Percentage (%)

Represents the Confidence Score, indicating the AI's certainty level for each detection.



Optimizing data flow for immediate action and long-term analysis

1

Urgent Situations

Immediate alerts sent to the management bureau for critical hazards.

High-priority road damage is flagged and transmitted in real-time to enable rapid response.

2

Routine Monitoring

Small cracks are logged and analyzed during scheduled routines.

Low-priority data is batched and processed during off-peak times for comprehensive analysis.

Benefit

Reduces bandwidth costs by prioritizing high-risk data while maintaining comprehensive monitoring coverage.

**Invisible to Visible,
Value to Revenue!**

Monetization Strategy

Subscription Model

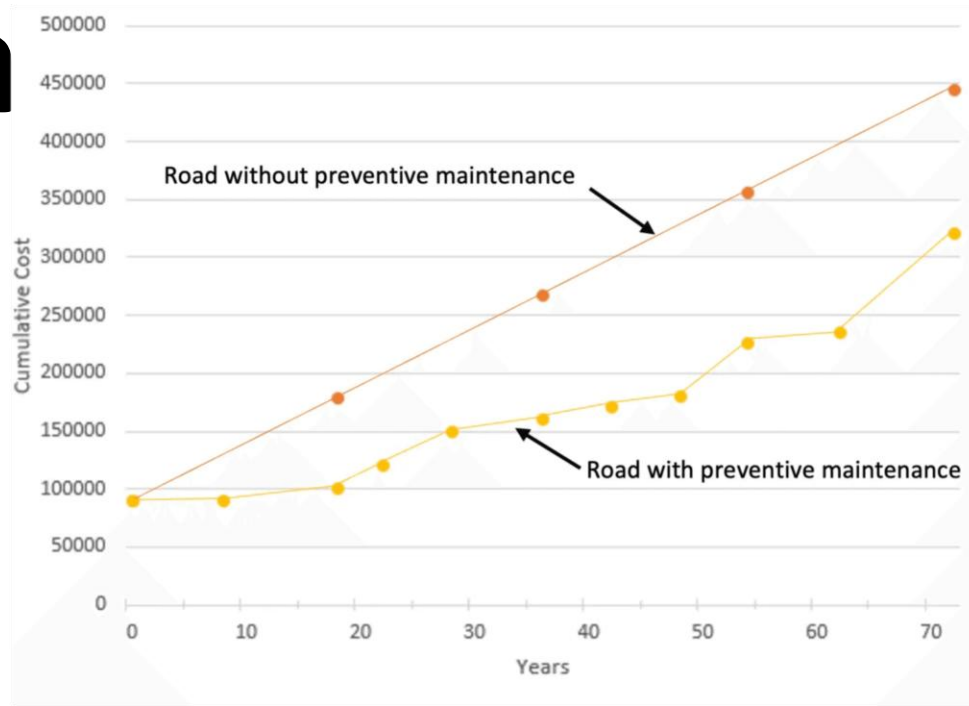
Cities and insurers pay for access to real-time road data and actionable insights.

Value for Cities

Saves millions in maintenance costs through early detection and preventive action.

Value for Insurers

Reduces claims by improving overall road safety and enabling data-driven risk assessment.



Our Teammates

Juhwang Park
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Project Management
Perception / AI

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AI Designer

Lea Zou



Data collector

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UI Designer

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Make the Invisible, Visible.

Team HELIOBOX